

MT21 : Optimization Techniques				
Teaching Scheme: Theory Session: Total 45 Hours		Credit: 03	Examination Scheme: Internal (TH): 25 Marks External (TH): 50 Marks Total :75 Marks	
Prerequisites: Basic mathematical knowledge is essential.				
Course Objectives: • To understand the role and principles of optimization techniques in business world. • To understand the process of problem statement formulation of the business scenario. • To understand the implementation of various decision-making techniques in the process of decision making. • To gain the techniques and skills on how to use optimization techniques to support the decision making in business world.				
Course Outcomes: On completion of the course, learners should be able to				
CO#	Cognitive Domain	Course Outcomes		
CO1	Apply	Understand and formulate linear programming models to solve optimization problems in various business contexts.		
CO2	Apply	Apply sequential models to make informed decisions in dynamic and uncertain environments.		
CO3	Apply	Utilize Markov chains and simulation techniques to model and solve complex inventory management problems.		
CO4	Apply	Apply PERT/CPM techniques to plan, schedule, and control projects effectively, including managing replacement decisions.		
CO5	Apply	Apply decision-making processes and strategic interactions using decision theory and game theory frameworks.		
Unit No.	Contents		Weightage in %	No of Sessions
1	Linear Programming 1.1. Various definitions, statements of basic theorems and properties, Advantages and Limitations 1.2. Application areas of Linear programming 1.3. Linear Programming – Concept 1.4. Simplex Method and Problems (No Graphical Solutions) 1.5 Transportation Problem (NWCM, LCM, VAM) optimize the problem using MODI Method		20	12
*Mapping of Course Outcomes for Unit 1: CO1				
2	Sequential model, Queuing Theory and related Problems 2.1 Processing n jobs through 1 machine, 2 machines and 3 machines		20	05

	Queuing Theory 2.2 Characteristics of Queuing Models, Transient and Steady states of the System 2.3 Model – I [(M/M/1) : (FCFS / ∞ / ∞)] Miscellaneous Problems based on above		
*Mapping of Course Outcomes for Unit 2: CO2			
3	Markov Chains, Simulation Techniques & Inventory Theory Markov Chains, Simulation Techniques 3.1 Markov chains: Applications related to technical functional areas, 3.2 Steady state Probabilities and its implications, 3.3 Decision making based on the inferences *Miscellaneous Problems based on above	20	08
*Mapping of Course Outcomes for Unit 3: CO3			
4	PERT CPM 4.1 Basic differences between PERT and CPM. 4.2 Network diagram 4.3 Time estimates (Forward Pass Computation, Backward Pass Computation), Critical Path 4.4 Probability of meeting scheduled date of completion, 4.5 Calculation on CPM network. Various floats for activities Event Slack 4.6 Calculation on PERT network. Miscellaneous Problems based on above	20	10
*Mapping of Course Outcomes for Unit 4: CO4			
5	Decision Theory & Game Theory Decision Theory 5.1 Introduction and Steps of Decision-Making Process 5.2 Types of Decision-Making Environments 5.3 Decision-Making Under Uncertainty 5.4 Decision-Making Under Risk Game Theory 5.5 Introduction, n X m zero sum game with dominance 5.6 Solution using Algebraic, Arithmetic and Matrix strategy	20	10
*Mapping of Course Outcomes for Unit 5: CO5			
Learning Resources			
Recommended Books: Text Books: - Operations Research by Pannerselvam - Operations Research Theory and Application by J. K. Sharma –Mac-MillanPublication			

- Statistical and Quantative Methods – Mr. Ranjit Chitale

Reference Books:

- Statistical Methods – S.P.Gupta, Sultan Chand, New Delhi
- Operation Research by V. k. Kapoor
- Operations Research by Kanti Swaroop, P. K. Gupta and Man Mohan
- Introduction to Operations Research by Hiller & Lieberman, Tata Mc GrawHill
- Operations Research by H. A. Taha
- Operation Research by Hira & Gupta
- What is Game Theory?, David K. Levine, Economics, UCLA

Research Software:

- MS Excel Solver
- TORA
- Python and / or R programming

Websites:

- www.orsl.in
- www.atzoperationalresearch.com

Recommended Certifications:

- Data science with R programing
- Certification in Tableau